# Yoga Pose Recognition and Angle-Based Grading System

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**ABSTRACT**

This paper presents a real-time yoga pose recognition and system that integrates skeleton-based pose estimation with deep learning and angle-based scoring. Traditional yoga guidance often relies on the subjective judgment of instructors, which may be inconsistent and inaccessible for remote users. The proposed system combines a keypoint detection framework with a lightweight convolutional neural network to classify ten common yoga poses. An interpretable scoring mechanism evaluates joint angle deviations by comparing user performance against standard pose templates. The system requires only a single webcam and provides real-time visual feedback, including annotated video frames and performance reports. Experimental results demonstrate high classification accuracy and consistent scoring, highlighting the system's effectiveness in delivering actionable feedback for home-based or remote yoga practitioners.

## 1. INTRODUCTION

In recent years, public awareness of health and fitness has grown significantly, with yoga becoming an increasingly popular method for both physical training and rehabilitation. However, the benefits of yoga depend heavily on the precision of each pose; incorrect postures not only reduce the intended effect but may also cause physical harm. Conventional yoga instruction, which relies heavily on in-person guidance from instructors or therapists, tends to be subjective, inconsistent, and resource-intensive. This makes it challenging to apply in large classes or remote learning environments. To overcome these limitations, this research proposes a technological solution that enables automated and accurate evaluation of yoga postures.

Despite the popularity of yoga, affordable and effective tools for home practice remain limited. Many existing systems require costly hardware or merely perform basic pose classification without offering actionable feedback on posture accuracy. This project is driven by the need to create a system capable of providing users with quantitative, real-time feedback, helping them assess the correctness of their poses and identify areas for improvement.

This study aims at developing a system capable of automatically classifying the user's yoga pose using a pre-trained classification model. It will also calculate and analyze specific joint angles based on a skeletal model. By comparing these real-time angles with standard pose references, the system will generate a detailed score reflecting the accuracy of the user's posture. Furthermore, it will provide visual and numerical feedback, including skeleton overlays and evaluations, and generate a performance summary report for further review or guidance

## 2. METHODOLOGY

### 2.1. Framework Pipeline

The overall system architecture consists of six key modules: video input preprocessing, keypoint detection, pose classification, standard pose referencing, angle-based evaluation, and annotated result output. These modules work sequentially to process user-uploaded yoga videos and provide both visual and quantitative feedback. Users begin by placing pre-recorded yoga practice videos into a designated input folder. The system then automatically reads each file, extracts video frames, performs format conversion if necessary, and prepares the frames for analysis. This preprocessing ensures consistent input quality across different video sources.

For keypoint detection, we adopt Google's MediaPipe Pose framework [2], which uses deep learning to detect 33 anatomical landmarks, including the head, shoulders, elbows, hips, knees, and ankles for each frame. MediaPipe operates in real time on standard CPUs, and its robustness to occlusion and movement makes it suitable for use in uncontrolled home environments. The extracted skeletal coordinates serve as the foundation for downstream classification and evaluation.

To recognize the yoga pose being performed, we trained a deep learning model based on the ResNet18 architecture [5], a residual neural network known for fast convergence and competitive accuracy. The training data was drawn from the Yoga-82 dataset [4], a benchmark dataset

containing over 28,000 pose images. We selected 10 commonly practiced yoga poses and created a focused subset. To improve generalization, data augmentation techniques such as rotation and horizontal flipping were applied. In addition to the Yoga-82 subset, we collected around 200 additional images per pose for fine-tuning. Frames are converted into tensors and normalized before being passed to the model, which then classifies the current pose.

After classification, the system conducts joint angle analysis to evaluate pose accuracy. Standard reference angles for each pose were compiled from the widely used online resource "Yoga with Adriene"[12]. For each pose, the 33 keypoints were converted into a set of ideal angles, stored in CSV files. During evaluation, the system calculates actual joint angles using a three-point definition and the law of cosines [1]. This allows the system to measure, for instance, the knee angle using the coordinates of the hip, knee, and ankle.

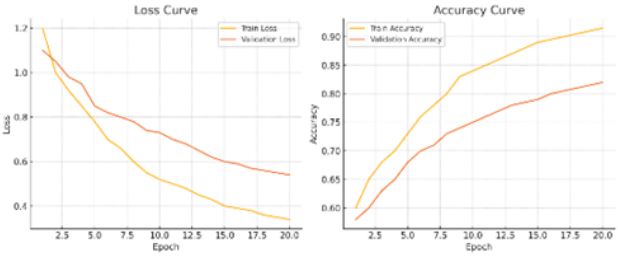
To interpret these differences, we adopt a nonlinear scoring strategy applies gentle penalties to minor deviations and stronger penalties to larger misalignments. Each angle receives an individual score, and the average of these scores is used to quantify the frame-level pose quality. This approach allows for accurate, interpretable feedback on specific body parts and supports personalized correction.

Finally, the system overlays real-time annotations onto the video, including the predicted pose name, each joint's measured angle, its deviation from the standard, and a color-coded skeleton rendering. In parallel, a CSV report is generated that records frame-by-frame classifications, joint angles, and corresponding scores. This dual-output format enables users to receive instant visual guidance during practice and perform detailed post-session analysis. Our result logging design is adapted from Sharma et al. [6], who proposed a frame-level feedback framework that stores pose recognition results and quantitative metrics for each frame to support progressive evaluation in fitness training.

### 2.2. Overview of Yoga Poses

To ensure the feasibility and manageable scope of this To ensure the feasibility and manageable scope of this research, we selected ten common and representative yoga poses as the primary targets for classification and evaluation. These include Bridge Pose, Downward Dog, Cobra, Extended Side Angle, Fish Pose, Tree Pose, Warrior I, Warrior II, Warrior III, and Cat-Cow. This set covers a diverse range of difficulty levels, body dynamics, and limb configurations, providing comprehensive test cases for evaluating the system's pose recognition accuracy and scoring performance.

### 2.3. Models




(a) (b)

Fig. 1. Training and validation loss/accuracy curves for ResNet18, (a) Loss curve, (b) Accuracy curve.

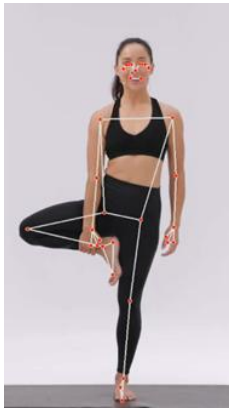


Fig. 2. Visualization of annotated keypoints.

To evaluate the classification model's performance, we trained a ResNet18 convolutional neural network using a curated subset of the Yoga-82 dataset. The training process spanned 20 epochs, during which both training and validation loss consistently decreased, while classification accuracy steadily improved. The model achieved approximately 91.5% training accuracy and 82% validation accuracy. These results confirm the model's effective generalization capability for yoga pose recognition.

Fig. 1 presents the training and validation loss and accuracy curves. The left subplot shows the downward trend in loss, while the right subplot displays the accuracy improvement, indicating convergence and learning stability.

To provide users with intuitive visual feedback, we utilized OpenCV to annotate each video frame with pose-related overlays. The system draws joint keypoints and connecting skeleton lines in real time, aiding the user in understanding their posture alignment during each frame. An example of this visual output is shown in Fig. 2, where the detected landmarks and skeleton structure are rendered directly onto the image.

### 2.4 Grading Logic Design

**Table 2.** Scoring Output Standard Table

| Error Range (°) | Score Formula | Score Interval |
|---|---|---|
| ≤ 20° | 100 - 0.5 × error | 90–100 |
| 20°- 40° | 90 - 1 × (error − 20) | 70–90 |
| 40°- 60° | 70 - 2 × (error − 40) | 30–70 |
| >60° | Score = 0 | 0 |

In addition to pose classification, the system integrates a quantifiable scoring mechanism to assess the quality of each yoga pose. The scoring logic is designed to account for natural body variation and measurement tolerance, avoiding excessive penalties for small deviations. Each joint angle is individually evaluated and assigned a score based on its difference from expert-defined reference angles. The scoring mechanism follows a nonlinear penalty structure in which larger errors are penalized more harshly while minor deviations incur minimal deductions. This enhances scoring robustness and better reflects realistic posture assessment. The specific scoring formula and error intervals are detailed in Table 2, which defines how different angular deviations map to numerical scores.

To compute these joint angles, the system utilizes a three-point geometric method based on the law of arccosines [1], where the vertex point (B) and its two neighbors (A and C) define the angle. This method provides accurate angular estimates even when slight variations exist in keypoint detection. The formula used is:

$$\theta = \arccos\left(\frac{\vec{BA} \cdot \vec{BC}}{||\vec{BA}|| \cdot ||\vec{BC}||}\right)$$

Here, vectors BA and BC are formed from three body keypoints, and point B is the joint at which the angle is measured. This approach allows the system to evaluate the precision and completeness of each body segment's posture.

Based on the structural characteristics of each pose, we defined four key angle features that form the core of our evaluation logic: body_line_angle, knee_angle, bent_leg_angle, and shoulder_angle. These angles reflect stability zones and limb positioning critical to posture execution. For example, Warrior I is assessed by combining knee_angle, bent_leg_angle, and shoulder_angle to verify proper front leg flexion and arm extension.

The standard reference angles used for comparison were compiled from expert-led instructional videos by Adriene Mishler [11]. Once a pose is classified, the system retrieves the corresponding reference angles and calculates the error with respect to the user's joint angles.

At runtime, the system scores each video frame individually. It overlays on-screen elements such as the predicted pose label, average pose score, per-joint angle measurements, and visual deviation indicators. Simultaneously, it logs structured data into a CSV file that records joint angles, scoring components, and pose classification results.

This real-time dual-output design is inspired by smart training frameworks introduced by Aggarwal et al. [12] and Dutta et al. [9], which emphasize combining visual overlays (e.g., skeleton lines, angle annotations) with quantitative metrics to enhance user awareness, enable real-time self-correction, and support post-session analysis.

## 3. SYSTEM IMPLEMENTATION AND RESULTS

### 3.1 System Output Demonstration




(a) (b)

Fig.3. Output Sample 1



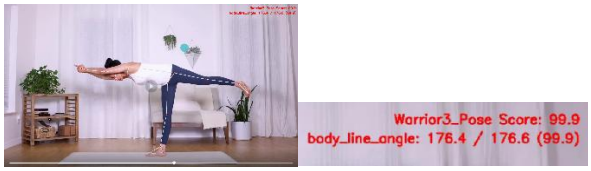

(a) (b)

Fig.4.Output Sample 2




(a) (b)

Fig.5.Output Sample 3




(a) (b)

Fig.6. Output Sample 4

Figs. 3–6 illustrate how the same pose can receive different scores, with corresponding visual differences in annotated feedback. From these examples, we observe that the system may initially misclassify transitional movements as other poses, but ultimately corrects itself once the pose is

stabilized.

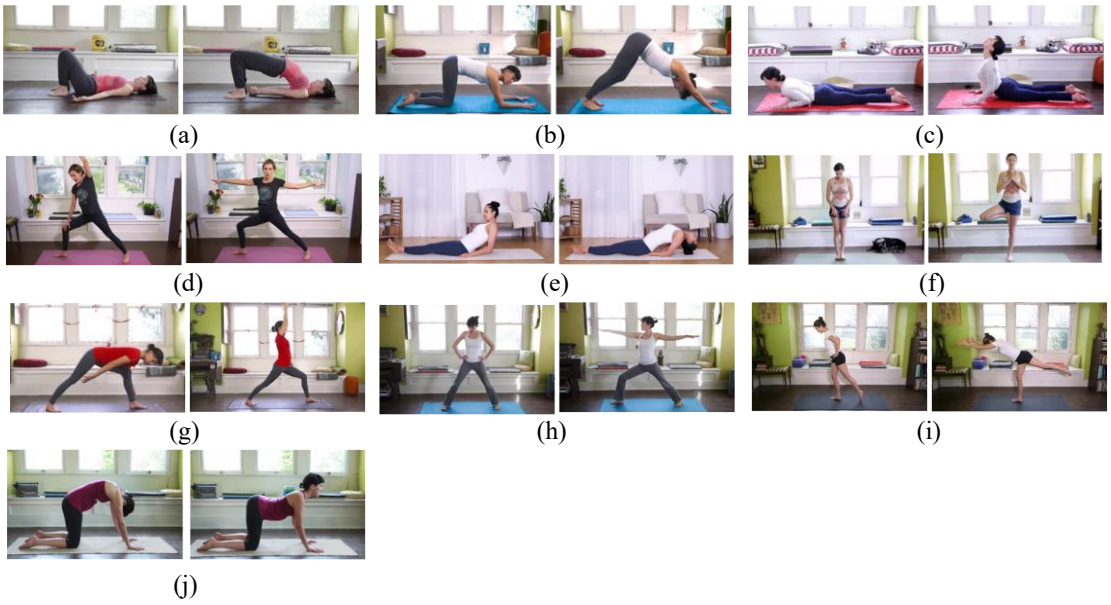



Fig.6. Representative frames for all evaluated yoga poses, (a) Bridge pose, (b) Downward dog, (c) Cobra, (d) Extended side angle,(e) Fish pose, (f) Tree, (g) Warrior I, (h) Warrior II, (i) Warrior III, (j) Cat cow

### 3.2 Segment-Level Classification Accuracy

Table 3. Confusion Matrix Table

| Ground Truth \ Predicted | Bridge | Down Dog | Ext Side Angle | Fish | Tree | Warrior I | Warrior II | Warrior III | Cat-Cow | Cobra |
|---|---|---|---|---|---|---|---|---|---|---|
| Bridge | 981 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Down Dog | 0 | 312 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Ext Side Angle | 0 | 0 | 410 | 0 | 909 | 0 | 0 | 0 | 0 | 0 |
| Fish | 0 | 0 | 0 | 877 | 0 | 0 | 0 | 0 | 0 | 0 |
| Tree | 0 | 0 | 0 | 0 | 1021 | 0 | 0 | 0 | 0 | 0 |
| Warrior I | 0 | 0 | 0 | 0 | 300 | 1100 | 0 | 0 | 0 | 0 |
| Warrior II | 0 | 0 | 0 | 0 | 50 | 0 | 800 | 0 | 0 | 0 |
| Warrior III | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 650 | 0 | 0 |
| Cat-Cow | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 430 | 10 |
| Cobra | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 5 | 620 |

To improve classification robustness and reduce frame-level prediction noise, we adopted a segment-level majority voting strategy. Each video segment—corresponding to a complete yoga pose—was assigned the pose label that appeared most frequently across all frames. Additionally, predicted labels were standardized via a mapping scheme to ensure consistency with ground truth annotations. As shown in Table 3, this approach achieved a segment-level classification accuracy of 85.1% across ten common yoga poses. The evaluation dataset consisted of 8,465 annotated frames, with Ground Truth pose counts also totaling 8,465, distributed unevenly across poses depending on video length. Individual segments ranged from 312 to 1,400 frames, depending on pose duration. This result demonstrates that majority voting and label normalization are effective for improving classification stability in real-world yoga practice scenarios.

To provide a visual reference for the evaluated poses listed in Table 3, Fig. 6 presents example frames for each yoga posture. Each pose includes three representative frames—typically taken from the beginning, middle, and end of the movement—to illustrate dynamic variation and scoring context.

## 4. SYSTEM FEATURES AND INNOVATIONS

The proposed system offers several key features that enhance its practicality, flexibility, and scalability in real-world yoga pose analysis and feedback scenarios.

One notable design decision is the integration of full RGB image input for pose classification, instead of relying solely on keypoint coordinates. By feeding image data into the ResNet18 model [5], the system can interpret spatial and contextual cues such as body orientation, camera perspective, and occlusion. This approach significantly improves classification robustness, especially in cases where skeleton-only input may lead to ambiguity—for example, when distinguishing between mirrored poses.

In terms of scoring analysis, the system maintains detailed per-frame records, including pose classification results and angle-based evaluations. Each frame is associated with an index, a predicted pose label, the measured and standard joint angles, and their corresponding scores. A final frame-level score—typically an average across multiple joints—is also computed. These data are exported in structured CSV files, facilitating real-time review and long-term tracking. This traceable scoring strategy builds upon the motion quality evaluation framework proposed by Wang et al. [3], which introduces a nonlinear, tiered scoring mechanism to distinguish between minor and major joint

angle deviations, enabling more accurate assessment of movement quality across time.

This example presents a scoring chart from a session involving the Bridge Pose. The user's angles gradually aligned with the reference standards, resulting in a smooth increase in score up to 97.3. This visual progression reflects the system's ability to recognize subtle pose refinements and provide meaningful feedback for improvement.

Another advantage of the system is its modular scoring mechanism. Because the scoring module is decoupled from the classification model, it can be updated or extended without retraining the neural network. Scoring thresholds and reference angles are stored in editable CSV files, allowing new poses to be added via simple data entry. The system can also support user-specific customization, enabling personalized evaluation standards. This design significantly enhances adaptability across different use cases [13].

Finally, the system operates as a fully automated, end-to-end pipeline. From video input to output visualization and CSV export, all processes—including frame extraction, pose detection [2], classification [5], angle evaluation [1], visual overlay rendering [6], and results logging—are executed without manual intervention. This complete automation enables consistent operation and makes the system especially suitable for app-based deployments, remote learning, and unsupervised fitness practice.

## 5. CHALLENGE AND BREAKTHROUGHS

Throughout the development and evaluation of the system, we encountered several technical challenges related to pose estimation, movement scoring, and classification consistency. These issues were addressed through iterative experimentation, algorithm design, and architectural refinements. The following highlights three key challenges and the strategies we employed to overcome them.

One major challenge involved selecting an appropriate keypoint detection engine. We compared OpenPose [10], HRNet [11], and MediaPipe Pose [2] in terms of inference speed, robustness, and integration feasibility. Although OpenPose and HRNet offer high-accuracy detection, they require GPU acceleration and are computationally expensive, resulting in delays during inference. HRNet, in particular, demands CUDA-enabled environments, which limits portability. In contrast, MediaPipe Pose provides 33 anatomical landmarks at over 25 FPS on standard CPUs, with good stability under occlusion. It also offers a well-documented Python API and built-in visualization support, making it the most practical and accessible choice for a lightweight, real-time yoga evaluation system.

Another obstacle arose from the initial version of the scoring function. The first design implemented a simple subtraction-based method, calculating scores by directly deducting angular deviation from 100. However, this naive approach consistently produced overly high scores (often above 90), failing to capture the subtle differences in pose quality. To improve discriminative power, we adopted a nonlinear penalty scheme, in which small angular errors are tolerated, while larger deviations are penalized more steeply. This tiered scoring system divides the error range into tolerant, warning, alert, and error zones, each with customized penalty slopes. Combined with structured CSV output, the system now offers clear, actionable feedback for both real-time corrections and long-term training analysis.

Lastly, we observed consistent misclassification errors arising from pose symmetry and perspective distortion. For symmetric poses such as Tree Pose and Warrior I/II, left-right variations were often confused due to their mirrored spatial configurations. To resolve this, we applied horizontal flip augmentation during training and treated mirrored examples as equivalent, guiding the model to learn semantic identity over raw spatial layout. This approach improved generalization for directionally ambiguous data. Additionally, camera distance and scaling discrepancies introduced further classification noise, as body proportions varied with the user's position. To mitigate this, we implemented pose normalization using anchor-based scaling (e.g., pelvis center, shoulder width), aligning all samples to a common frame of reference. Similar normalization strategies have also been employed successfully by Dutta et al. [8].

Together, these refinements improved model accuracy, reduced false positives, and enhanced the system's robustness to user variability and recording environments—critical traits for real-world deployment.

## 6. CONCLUSION AND FUTURE WORK

### 6.1 Conclusion

We developed a real-time yoga scoring system that integrates MediaPipe Pose for skeletal detection, ResNet18 for pose classification, and a nonlinear scoring method based on joint angles. The system delivers accurate classification, interpretable feedback, and modular design—all operating on standard hardware without the need for additional sensors. Its flexibility allows for easy expansion with new poses or personalized scoring standards. Overall, the system provides an efficient, low-cost solution for home-based yoga practice, remote instruction, and physical rehabilitation.

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